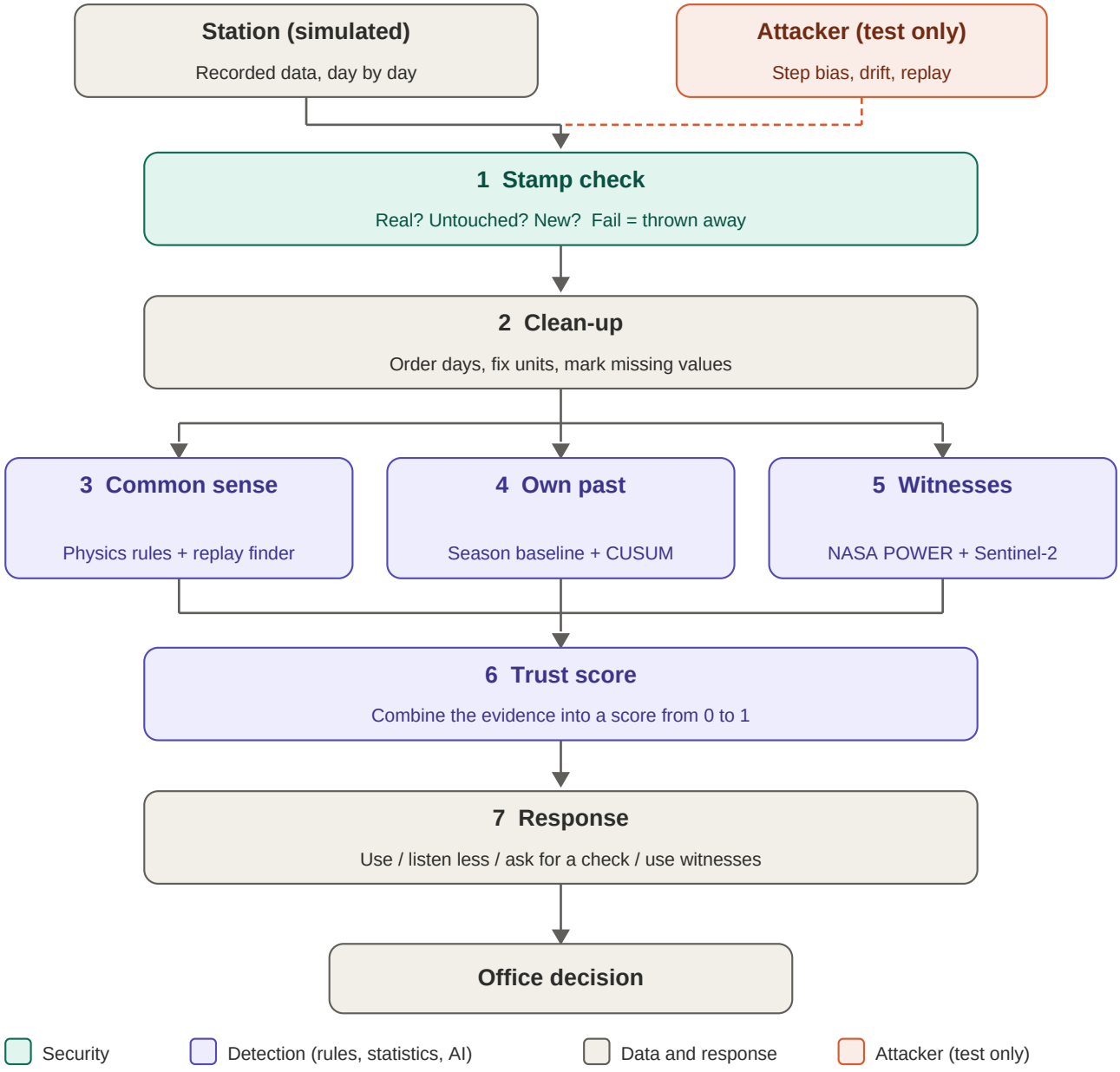
The Guard

Architecture to protect weather-station data from false numbers (TSYP 14, Track 2: False Data Injection)

A **guard** stands between the weather stations and the office. Before a number is used, it passes **7 layers**. Layer 1 is **security**. Layers 3 to 6 are **detection** (rules, statistics and AI). Layer 7 decides what the office should do.



Threat model in 3 lines
Attacker: changes numbers on their way to the office, sends old numbers again, or makes a hacked station send slightly wrong numbers.
We protect: the office's decision (water, drought, insurance payment).
We assume: a station's secret key is not stolen. If the station itself is hacked, layers 3, 4 and 5 catch the lies.

The layers (1 to 6)

1 **Stamp check** SECURITY (trust anchor)	**Asks:** Did this really come from the station, untouched, and is it new? **Use: Ed25519 signature** on each day's record (Python library *cryptography*), plus the date, a **counter** that goes up by 1 every day, and a **hash chain** (each day carries the fingerprint of the day before). Lighter option: **HMAC-SHA256** (stamp made with a shared secret). **Result:** stamp fails = number thrown away and alert. **Limit:** it cannot catch a hacked station that stamps lies. That is why layers 3 to 5 exist.
2 **Clean-up** DATA PREPARATION	**Asks:** Is the data tidy and comparable? **Use: pandas**: put days in order, convert units (degrees C or Kelvin), mark missing values as missing, keep the full history. **Why:** wrong units or fake zeros would look like attacks.
3 **Common sense** RULES	**Asks:** Is this physically possible? Did this week already happen? **Use:** simple rules in **pandas**: minimum <= mean <= maximum, humidity between 0 and 100%, dew point <= temperature, no negative rain or wind, sunshine <= clear-sky value. **Replay finder:** compare each 7-day block with all earlier blocks. A match = replay. **Catches:** impossible numbers and replayed weeks.
4 **Own past** STATISTICS	**Asks:** Is today normal for this station at this time of year? **Use:** (1) **seasonal baseline** = typical value of each day of the year from past years (median and spread); (2) **z-score** of today's gap; (3) **CUSUM** = running total of small gaps in the same direction; (4) **ruptures** library to find the day the average jumped. **Catches:** step bias (12%) and slow drift. **Upgrade:** predict today's value from the station's other variables (**regression**) to remove natural ups and downs.
5 **Witnesses** STATISTICS	**Asks:** Do independent sources agree with the station? **Use: NASA POWER**: gap = station minus POWER every day; learn the normal gap on clean data; run CUSUM on the gap. **Sentinel-2**: **cloud mask** and **NDVI** (how green the ground is). Example: strong sunshine reading but a cloudy image = suspicious. No image that day = "no evidence", not "disagree". **Catches:** step bias, drift, and replay (old days do not match today's weather). **Note:** POWER is an area estimate, so small gaps are normal. Wait for repeated gaps.
6 **Trust score** AI + STATISTICS	**Asks:** Overall, how much do we believe this station today? **Use: logistic regression** (scikit-learn) turns the scores of layers 3, 4 and 5 into a **compromise probability** p. Optional extra score: **Isolation Forest** (PyOD) for odd patterns. Trust moves like a battery: **trust = 0.9 x yesterday + 0.1 x (1 - p)** (example, tune it). **Trained on:** validation data where we injected attacks ourselves.

Layer 7 and the test lab

7 **Response** DATA AND ACTION	**Asks:** What should the office do with this number? **Use:** a small table of actions (below). The limits are examples: set them on validation data.

Trust score	What the guard does
0.8 to 1	**Use as is.**
0.6 to 0.8	**Listen less (de-weighting):** value = trust x station + (1 - trust) x witness.
0.3 to 0.6	**Ask for a check** (verification request) and raise an alert.
below 0.3	**Quarantine:** use the witnesses only.
Stamp fails	**Throw the number away** and raise an alert.

Test lab (outside the guard): we play the attacker

Attack (official standard test)	How we make it
Step bias 12%	From a chosen day, multiply the values by 1.12.
Drift up to 15%	The error grows from 0 to 15% over several weeks.
Replay of 7 samples	Copy 7 old days over 7 new days.

Replay the data day by day: the guard never sees tomorrow.
Hide-and-seek attacks (to test evasion): a smaller bias (6%), a slower drift, a replay of a similar-season week, an attack on cloudy days, an attacker who knows your limits.
Hidden site test: train on some sites, test on a new one (**leave-one-site-out**), 5 to 10 sites in different climates.
Fair split: tune on early years, final test on later years.
We measure: detection rate, false alarms per 100 days, detection delay in days, memory, cost of the stamps (time and size), and wrong decisions **with** and **without** the guard.

Tools and hardware

Software: Python 3.10+, pandas, numpy, scikit-learn (logistic regression, Isolation Forest), ruptures, PyOD, cryptography, requests (NASA POWER API). Copernicus Browser for Sentinel-2.
Hardware: any laptop (4 cores, 8 GB RAM), no GPU, nothing to buy. The organizers say running on a laptop is fully valid.
Size: one row per day, so it should run in seconds. Measure it and report it.

What the jury asks for, and where we answer

The 6 deliverables	Where in this architecture
1. Threat model	Threat model box (page 1) and the test-lab attacks.
2. Architecture and trust anchor	The diagram and layer 1 (stamp check).
3. Signal processing	Layers 2, 4 and 5 (clean-up, time series, change points, context).
4. AI or statistical detection	Layers 3 to 6, ending in the trust score.
5. Adversarial validation and response	Layer 7 and the test lab.
6. Documentation with official data	NASA POWER and Sentinel-2 L2A, official standard test.

3 questions to ask the organizers
1. Which variable is the "real series", which source is the station and which is the reference? 12% of what (degrees C or Kelvin)?
2. Which Sentinel-2 indicators count as context (cloud mask, NDVI, others)?
3. How are the metrics defined (delay in days, false alarms per what)?

Technical words, in easy words

Trust anchor	The secret-key part that proves where the data came from.
Signature	A stamp only the station can make. Change one digit and it breaks.
Hash, hash chain	A short fingerprint of data. In a chain, each day carries yesterday's fingerprint.
Counter	A number that goes up by 1 each day. A repeated or skipped number is suspicious.
Replay	Sending old real numbers again.
Step bias	Numbers suddenly too high or too low by a fixed percent.
Drift	An error that grows a little more each day.
Baseline	What is normal for this place on this date of the year.
z-score	How many usual ups and downs a number is away from normal.
CUSUM	A running total of small gaps in the same direction. One gap is chance; many in a row is not.
Change point	The day when the average of a series jumps.
Regression	A model that predicts one number from other numbers.
Logistic regression	A small model that turns several scores into one probability.
Isolation Forest	A model that finds odd points, because they are easy to separate from the crowd.
De-weighting	Listening to a source less.
False alarm	Normal data wrongly flagged as an attack.
Leave-one-site-out	Train on some sites, then test on a site the guard has never seen.
NASA POWER	A NASA service with daily weather for any place on Earth.
Sentinel-2	A European satellite that photographs the ground every few days.
NDVI, cloud mask	NDVI = how green the ground is. Cloud mask = which parts of the image are clouds.
Clear-sky radiation	The most sunshine possible when there are no clouds.